Booking Backend → AI Integration

Handoff for the booking teammate • Prepared 22 October 2026

1. What already works

The AI service is connected to **POST /internal/ai/availability**. The backend builds and starts. The AI Docker container can reach the backend on port 3000. Live Gemini chat and demo property search have passed.

A live booking-tool request reached the backend and returned **404: property not found**. The AI correctly reported that error. This verifies connectivity and error handling; it does **not** verify successful slot retrieval or booking creation.

All 14 AI automated tests passed inside Docker. Those tests use simulated dependencies. Property recommendations still use a demo catalog because the real property search/detail endpoints are not present in the inspected main revision.

2. First priority: reproducible database setup

Problem: the current Docker initialization file, `database/schema.sql`, creates the older bookings/calendar_events tables. It does not create the tables expected by the new Prisma booking models:

- AgentAvailability
- CalendarException
- Visit
- VisitStatusHistory

Please provide: a reviewed migration or documented development setup that creates these tables and their relationships. Include exact commands, working directory, and environment requirements. Explain how to update an existing database without deleting its data. Restarting PostgreSQL alone does not apply schema changes to an existing volume.

3. Add a small, repeatable test dataset

The local database tested had **0 properties and 0 users**. Provide seed data or a supported setup flow containing:

- One organization and one owner/agent in that organization.
- One real property belonging to that organization.
- Agent availability on a known date.
- A visit and an exception that block some time, to verify conflicts.
- Lead/contact data needed for booking creation, coordinated with the CRM owner.

Send back: the property UUID, a date with available slots, expected slots, and the setup command. The current endpoint selects the organization’s first owner; confirm whether that is the intended assignment rule. Demo IDs such as `prop-1` are not valid booking IDs.

Availability Endpoint

Existing endpoint: POST /internal/ai/availability

4. Keep the agreed request and response shape

```
Request
{
  "property_id": "<real-property-uuid>",
  "preferred_date": "2026-11-01",
  "timezone": "Africa/Casablanca"
}

Success (example only; these are not verified real slots)
{
  "property_id": "<same-property-uuid>",
  "date": "2026-11-01",
  "slots": [{
    "starts_at": "2026-11-01T10:00:00+01:00",
    "ends_at": "2026-11-01T10:30:00+01:00",
    "agent_id": "<agent-uuid>",
    "agent_name": "Test Agent"
  }]
}
```

Return `slots: []` when the lookup succeeds but no times are available. Keep failures distinct: for example, 404 for a missing property or agent. Use timezone-aware ISO timestamps; UTC timestamps with Z are also valid.

5. Fix the issues found during inspection

Timezone: the service accepts timezone but creates day boundaries using UTC. Interpret the requested day in the requested timezone, then convert boundaries to UTC for database comparisons. Validate the date and supported timezone; currently both are only checked as strings.

Overnight overlaps: visits and exceptions are fetched only if they start during the requested day. An event that starts the previous evening and ends today can be missed. Fetch intervals that overlap the day, then apply the slot conflict checks.

```
Overlap rule for a half-open interval [start, end):
event.start < range.end AND event.end > range.start
```

Use the same principle for availability intervals crossing midnight. Keep cancelled visits from blocking slots. Confirm whether availability rows represent individual visit slots or larger working windows; define visit duration if windows must be split.

Authentication and organization scope: the inspected controller has no guard and the property lookup uses only its ID. Coordinate with the auth owner to validate service credentials and restrict property/agent access to an authorized organization. Merely receiving an organization header is not sufficient authorization.

The AI can send `Authorization: Bearer <AI_SERVICE_TOKEN>` and `X-Organization-Id` from server configuration. Agree the credential and organization validation rules before multi-agency use.

Booking Creation & Acceptance Checklist

Next endpoint required: POST /internal/ai/visits

6. Implement real booking creation

This endpoint is missing in the inspected backend. Until it exists, the AI can check slots but cannot reserve or confirm a visit. The following payload follows the supplied team API contract; coordinate lead/contact identity with the CRM owner.

```
Request
{
  "lead_id": "<lead-uuid>",
  "contact_id": "<contact-uuid>",
  "property_id": "<property-uuid>",
  "agent_id": "<agent-uuid>",
  "starts_at": "2026-11-01T10:00:00+01:00",
  "customer_notes": "Customer prefers a morning visit"
}

Success
{
  "visit_id": "<visit-uuid>",
  "status": "confirmed",
  "property_id": "<property-uuid>",
  "agent": {"id": "<agent-uuid>", "name": "Test Agent"},
  "starts_at": "2026-11-01T10:00:00+01:00",
  "ends_at": "2026-11-01T10:30:00+01:00"
}

HTTP 409
{"error": "SLOT_UNAVAILABLE",
 "message": "This time slot is no longer available."}
```

Validate organization access, related IDs, and the selected slot again when saving. Make conflict checking and visit creation concurrency-safe: two requests for the same slot must not both succeed. Derive the end time on the backend. Keep calendar updates consistent with the booking, as required by the team contract.

7. Checks before handing it back to AI

1. A fresh setup and an existing-database upgrade both work.
2. A seeded property returns the expected slots; no available times returns [].
3. Missing property/agent and invalid date/timezone return clear errors.
4. Existing visits, exceptions, and overnight conflicts block the correct times.
5. Missing credentials and cross-organization requests are rejected.
6. Booking succeeds once; competing requests cannot double-book.
7. A booked slot no longer appears as available.

8. What to send to the AI teammate

Send the commit/PR, migration and seed commands, test property UUID/date, expected response, and agreed authentication instructions. Share secrets through a secure channel, not this PDF. The AI teammate will then test real slots and add the booking-creation tool once the endpoint and required CRM IDs are ready.

Evidence: backend revision inspected: main 6eaaa20, merged into ilallali. Sources: booking controller/service, availability DTO, Prisma schema, database/schema.sql, live local checks, and AI_Microservice_Team_API_Contract.pdf (sections 2, 5, 6). Backend source was not modified during these checks.